

A Systematic Review of Machine Learning Models Against MIMIC and Currency Demand Models on Estimating the Size of Informal Economy

Jinsy Panamkattu
Department of AI Engineering
Jönköping University
Jönköping, Sweden.
paji25oo@student.ju.se

Neethu Joseph Chully
Department of AI Engineering
Jönköping University
Jönköping, Sweden.
Chne25ai@student.ju.se

Abstract—The informal economy really remains a challenge in the modern world. It is extremely difficult to estimate its size. The biggest challenge is its hidden nature due to unregulated methodologies. The traditional econometric approaches like Multiple Indicators Multiple Causes (MIMIC) model and the Currency Demand Approach (CDA) could fail even though they long dominated sources. The main disadvantage of using the MIMIC and CDA is that it could only rely on restrictive assumptions than nonlinear and heterogeneous characteristics. From the recent advance artificial intelligence (AI) and machine learning (ML), there arises new possibilities for improving estimation accuracy and flexibility. This systematic literature review clearly examines whether and how the machine learning models enhance or outperform traditional econometric methods such as MIMIC and CDA, in estimating the size of informal economy. This systematic literature review follows the review protocol established by Kitchenham (2004). It also includes the structured search which was conducted on Scopus and Web of Science databases which is limited within the time period between 2010 and 2025. After the iteration procedure which includes screening and quality assessment, 15 papers were shortlisted for review. Data were extracted and analysed through a qualitative thematic framework. The result points out four primary mechanisms, predictive accuracy, non-linearity handling, spatial learning, and mixed-data integration, through which ML models enhance or outperform traditional methods.

Keywords—Informal economy, shadow economy, MIMIC model, currency demand approach, machine learning, artificial intelligence

I. INTRODUCTION

The informal economy refers to economic activities that fall outside the reach of government regulation, taxation, and formal oversight. No country is exempt; its presence ranges from street vendors and small-scale traders to undeclared cash transactions between employers and workers. Several structural conditions feed into it: bureaucratic barriers, heavy tax burdens, poverty, unemployment, weak governance, corruption, and exclusion from formal financial systems [4]. In many developing countries, the formal economy simply cannot generate jobs fast enough to absorb the available labour force.

Measuring the informal economy typically means expressing it as a share of Gross Domestic Product (GDP) or total employment. The irregular work arrangements and unreported incomes will get captured through the unregistered enterprises which in turn helps authorities. The variables like tax pressure, institutional capacity, corruption,

joblessness, and household income levels are the key combinations that enables these factors. Where taxes run high and institutions are fragile, the informal sector tends to be larger. Estimation methods fall into two broad categories: direct approaches, which rely on household or firm-level surveys, and indirect ones, which infer activity from currency circulation patterns or composite indicator models. Machine learning has entered this space as well, improving predictive fit, though estimates remain sharply sensitive to whichever methodology a researcher chooses [6].

This review asks a focused question: Do machine learning models actually improve on or outperform established econometric tools when it comes to estimating informal economy size? The two benchmarks under scrutiny are the MIMIC model and the CDA. Drawing on a systematic reading of existing research, the analysis maps out how these methods differ in structure, what variables each one prioritizes, and how country context shapes results. Beyond that comparison, the review works to pin down the strengths and blind spots of each approach, examine whether estimates produced under different frameworks can meaningfully be compared, and flag the empirical gaps that future research needs to address

II. BACKGROUND

Measuring the informal economy remains a persistent methodological challenge in economic research due to the hidden and unregulated nature of informal economic activity. Traditional econometric approaches, particularly the MIMIC model and CDA, have long dominated the literature; however, both methods rely on restrictive assumptions that may fail to capture the heterogeneous and nonlinear characteristics of informal economic activity across diverse country contexts [7]. With the growing availability of artificial intelligence (AI) and machine learning (ML) techniques, researchers have increasingly explored whether data-driven approaches such as Random Forests and Support Vector Regression (SVR) can provide more accurate and flexible estimation frameworks [5]. The earlier reviews such as traditional econometric methods or emerging ML applications in isolation that we examined definitely help in giving evidences to many fragmented methodologies datasets, and regions, with no comprehensive synthesis. These compares systematic ML approaches and evaluates their respective strengths and limitations. Also, it estimates the informal economy in the specific context applicable in the circumstances. These gapes motivate the current scenario in reviewing.

III. RESEARCH METHODOLOGY

A. Review Protocol

The review is based on Kitchenham's [16] three-phase protocol — planning, conducting, and reporting.

Planning involved clearly defining the research goals and deciding which studies would be included and which would be excluded. The conducting phase involved a literature search, record screening, quality checks, and data extraction from studies that met the criteria. Those findings then given to a qualitative analysis which leads to the review's conclusions. This structure improves the reliability of the research process.

B. Research Question

In accordance with Kitchenham's [16] SLR framework, this review was structured to investigate the following research question:

RQ: How do machine learning models enhance or outperform traditional econometric methods, particularly MIMIC and CDA, in estimating the size of the informal economy?

C. Data Sources and Search Strategy

Scopus and Web of Science (WoS), are the two academic databases used for identifying the relevant literature. These sources cover multi-disciplinary areas, well organized and indexed and support a trusted systematic literature review. The search was limited to publications from 2010 to 2025, reflecting the period during which ML methods began gaining meaningful acceptance in economic and social science research.

The search strategy was built around four primary keywords.

- 1) “informal economy size” AND “machine learning” for SCOPUS
- 2) “informal economy size” AND “MIMIC” for WoS
- 3) “informal economy size ” AND “currency demand approach” for WoS
- 4) “informal economy size” AND “machine learning” for WoS

The results from the four different search queries were collected from both databases, and any repeated entries were identified and deleted. The remaining records were screened by title and abstract for relevance to the research question. After this selection, 42 papers were kept for full-text review and evaluated against selection criteria like validation and direct relevance to the topic. After full-text assessment, 15 studies satisfied all inclusion criteria and were selected for the final analysis.

D. Inclusion and Exclusion Criteria

To obtain a manageable and reliable set of studies, a structured screening process was applied based on Kitchenham's systematic review guidelines [16]. The research question at the heart of this review served as the main reference point whenever a decision was made about whether a particular study should be included or excluded.

Well before any papers were looked at individually, a working set of criteria was put together to make those calls as consistent as possible. For a study to make the cut, it had to be both empirical and peer-reviewed, and it had to deal directly with at least one of three things: applying ML to the measurement of informal or shadow economic activity, comparing machine learning tools against econometric approaches like the MIMIC model or the CDA or pushing forward how unobserved economic activity gets measured through data-driven means. On top of that, the work itself had to be methodologically transparent, backed by real empirical findings, and genuinely relevant to what this review was trying to answer.

Anything that did not tie back clearly to estimating the scale of the informal economy was left out. That meant setting aside studies where ML was being used for economic problems with no connection to informality, where MIMIC or currency demand methods were applied in completely unrelated settings, or where the informal economy was only ever discussed in qualitative or policy terms with nothing empirical to show for it. Work that treated machine learning or econometric modelling purely on its own terms, without any real grounding in informal economy research, did not fit the scope of this review and was excluded accordingly.

1) Primary Selection:

The opening stage of selection was intentionally light, resting on just two basic conditions before anything else came into play:

- Papers had to be written in English
- Publication had to fall **within 2010 and 2025** — This period was chosen because it corresponds to the rise of machine learning and deep learning in economics and social sciences, as well as growing recognition of the limitations of MIMIC and CDA.

Studies that did not meet either criterion were removed at this stage to ensure that the later screening process was not slowed down by irrelevant material.

2) Secondary Selection

Studies passing the primary screening stage were subsequently assessed based on their titles and abstracts. This stage aimed to determine whether each study was substantively relevant to the research question. A record was retained if its title or abstract indicated a clear focus on at least one of the following:

- Work that puts ML tools to use in estimating, modelling, or predicting how large the informal or shadow economy actually is.
- Any kind of direct engagement or head-to-head comparison with traditional econometric tools — the MIMIC model and CDA in particular, in the context of measuring shadow economic activity.
- Efforts to develop or test data-driven ways of getting at the informal economy size, whether as a standalone approach or as something meant to sit alongside existing indirect estimation methods.

Studies whose titles and abstracts revealed an exclusive focus on unrelated econometric applications, general machine learning benchmarking outside economic contexts, or informal economy topics addressed through purely qualitative or policy-oriented lenses were excluded at this stage.

3) *Full-text Assessment*

Papers that made it through the second stage were then pulled up in full and gone through carefully. Each one was held up against the complete set of criteria, whether the empirical work held up, and how directly the study actually contributed to answering the question this review was built around. Three things were looked for in particular:

- Whether the study had a clearly laid out ML methodology that was genuinely put to work in estimating informal economic activity, not just mentioned in passing.
- Whether the empirical findings were substantial enough to be meaningfully set against what traditional approaches like MIMIC or CDA have historically produced.
- Whether enough methodological detail was provided to make data extraction and synthesis a realistic and reliable exercise.

Following full-text review, 15 studies satisfied all inclusion criteria and were selected for the final synthesis, representing the most methodologically robust and thematically relevant contributions to the field.

E. Study Quality Assessment

The quality of the included studies was evaluated by examining several key aspects relevant to comparative research between machine learning and traditional econometric methods for estimating the informal economy. Several things were looked at when going through each study. To begin with, how clearly the research objectives were spelt out mattered particularly whether the study was trying to get better predictive results, get to grips with nonlinear relationships, or sharpen the structural interpretation of informal economy estimates. Closely tied to this was the question of how transparently the methods were laid out, covering everything from the choice of model through to how the data was prepared and what assumptions the whole approach was resting on.

A fair amount of attention also went to the datasets themselves. Things like how many observations were included, how far back the data stretched, which countries were covered — Romania [2], Algeria [7], Vietnam [8], Peru [9], Nigeria [10], the Balkans [11], Sweden [6] among others — and how the authors dealt with gaps or noise in the macroeconomic indicators all fed into the assessment.

How the results were measured and reported was another consideration. For predictive models, metrics like RMSE, MAE, and R-squared were looked at; for MIMIC-based work, fit indices such as RMSEA, CFI, and SRMR were the relevant benchmarks; and for currency demand studies, residual diagnostics carried most of the weight. Finally, whether the work could realistically be reproduced was

factored in. This includes checking for details on parameter tuning and cross-validation in the ML studies [1], [2], [5], and for sensitivity checks in the traditional ones, such as varying the currency ratio baseline [15] or the zero-informality base year assumption [14].

The results were evaluated in depth and coherence. Especially it explains how different models or methods are explained. Also, how the researchers compared or how they resolved the differences in results. Example: MIMIC shows a declining in the informal sector while ML shows an increase [11]. It also assessed how extent the authors acknowledged their work limitations.

The extent to which authors acknowledged the limitations of their work was also assessed. This included recognition of MIMIC's identification problems and sensitivity to model specification [7], [10], currency demand assumptions about constant velocity of money [15] and the absence of non-monetary transactions [14], as well as machine learning issues such as overfitting in small samples [8], lack of causal interpretability [5], and data leakage between training and test sets [4].

Finally, the practical value of the findings for real-world applications was considered, such as policy design for tax administration, anti-corruption strategies [4], labour market regulation [3], and cross-border informal activity monitoring [6]. Together, these criteria enabled a consistent and meaningful assessment of the methodological strength of the studies included in this review, distinguishing those that merely applied techniques from those that provided robust, comparative evidence on when and how machine learning enhances or outperforms traditional econometric methods.

F. Data Extraction and Synthesis

A uniform strategy was used to systematically extract information from the 15 included studies. For each study, the following was recorded: who wrote it and when, which country or region the analysis was based in and what method was actually used to arrive at informal economy estimates. Methods were sorted into two broad categories: those rooted in traditional econometrics and those drawing on ML tools. Additionally, the datasets and their key indicators, such as tax burden, regulatory quality, GDP per capita, currency ratios, unemployment rates, and shadow economy proxy variables, were systematically recorded, along with the evaluation metrics used for validation (e.g., RMSE, MAE, R-squared, goodness-of-fit indices, residual diagnostics).

Pulling everything together for synthesis meant grouping the studies thematically rather than treating each one in isolation. Three broad clusters emerged naturally from the material:

1. **Traditional Structural Methods** — this group brought together studies built around MIMIC models [7], [8], [9], [10], [11] and those working within currency demand frameworks [12], [13], [14], [15].
2. **Machine Learning Enhancement Methods** — here the focus was on studies putting random forests, gradient boosting, support vector machines, and neural networks to work on informal economy estimation [1], [2], [3], [4].

3. Advanced Deep Learning and Spatial Methods – covering deep learning architectures [5] and graph-based or neighbour-aware algorithms that incorporate spatial dependencies across neighbouring countries [6].

In addition, several application domains were considered within the informal economy literature, such as cross-country comparative analysis [1], [11], single-country time-series estimation [2], [7], [8], [9], [10], [13], [14], [15], policy-oriented economic cost assessment [4], and spatial dependence analysis [6].

After that, patterns having to do with predictive performance, validation strategies, and methodological limitations were recognized and scrutinized. Key performance patterns included the consistently higher predictive accuracy of machine learning models over traditional MIMIC and CDA, particularly when handling non-linear relationships [1], [5] and high-dimensional indicator spaces [2]. Validation patterns revealed that traditional methods rely heavily on in-sample goodness-of-fit and sensitivity analyses [7], [10], whereas ML studies employed cross-validation, train-test splits, and out-of-sample forecasting exercises [1], [3], [5]. Limitations patterns included the lack of causal interpretability in deep learning models [5], the identification problems and specification sensitivity inherent to MIMIC [7], [10], the strong assumptions underlying currency demand approaches (e.g., constant velocity of money, absence of non-monetary informal transactions) [14], [15], and the risk of overfitting when ML models are applied to small-sample country data [8].

These reviews scrutinized different type of methods to estimate informal economy. It definitely explored the challenges by researchers and policymakers face while choosing methodologies between them. These understandings give the strengths and limitations of these rapidly expanding fields. Mainly in guiding future work toward hybrid models that could combine the interpretability of econometric methods. It will also help the predictive power of machine learning.

IV. RESULTS AND DISCUSSION

A. Quality Assessment Result

Quality assessment across the 15 selected studies pointed to a solid methodological base in the informal economy estimation literature. Out of these papers, 13 scored high or excellent, with clear research objectives, well-described methods, appropriate data choices, and defined evaluation metrics. Among the stronger studies were those using deep learning architectures to measure unregistered economic activity — reporting RMSE, R^2 , and MAE to back up model accuracy [3]; machine learning techniques tested against structural equation models for the Romanian shadow economy, with cross-validation and feature importance analysis bolstering methodological credibility [2]; and spatially-informed ML frameworks drawing on high-quality Swedish register data to infer informal economy size from neighboring country characteristics [6].

On the traditional econometric side, high-quality studies included a post-transition structural estimation of Poland's informal sector with transparent specification and sensitivity analysis [14], a panel econometric analysis of regulatory intensity as a driver of informality in developing countries using fixed-effects estimation [12], a multi-country MIMIC application tracing the informal economy across the Balkans from 1996 to 2021 [11], and a cash demand econometric model for Serbia incorporating time-series diagnostics and multiple robustness tests [15]. The studies addressing Vietnam's and Pakistan's, informal and underground economies received moderate ratings. This is because of mixed qualitative-quantitative approaches. Primarily the selection criteria used for these model selections were like limited evaluation metrics and insufficient integration of methodological elements. These were underspecified model selection criteria.

B. Research Question Result

The four primary mechanisms used in the ML models to estimate the size of the informal economy are below. These should enhance or outperform traditional econometric methods like MIMIC and CDA.

- 1) Handling Non-Linear and Complex Interactions: Not like the traditional MIMIC models, that would assume linear causal structures (e.g., [7]; [10]; [9]), machine learning algorithms (e.g., Random Forest, Neural Networks) will automatically capture non-linear relationships and high-dimensional interactions. These are among the causes and indicators of the shadow economy. These do not require pre-specified structural equations. Researches have shown that deep learning models can uncover hidden patterns in unregistered economy, that linear models may fail to detect [5] these patterns.

- 2) Superior Predictive Accuracy and Generalization: The traditional approach in currency demand relies on historical residual comparisons. It drops off from the assumptions about velocity of money and baseline periods. This weigh on ML Models, ensemble methods and support vector machines. It also validates the lower out-of-sample prediction errors. The size estimation of the informal economy across multiple countries [1] using ML algorithms can significantly reduce forecasting bias. The researches from Romania shows these very accurately. The behavioral shifts [2] and structural breaks through ML techniques outperform traditional time-series models.

- 3) Incorporating Spatial and Neighbour Effects: The Traditional MIMIC models simply analyze the economy separately or it will use simple cross-country comparison variables. However, [3] and [6] shows that ML methods, that would use graph-based and neighbour-aware algorithms are more efficient. It can directly model spatial dependencies. It is similar by estimating a country's informal economy based upon neighbouring countries' characteristics. This will outperform the MIMIC's inability to seamlessly incorporate relational data.

- 4) Integration of Mixed-Data and High-Frequency Indicators: The low frequency macroeconomic time series (e.g., GDP, tax revenue, currency ratios) were the methods used earlier by tradition. The ML models can be used to

integrate qualitative survey data [13]. It also can combine high-frequency proxy variables. Many research has shown combining econometric residual estimates with ML classification techniques. These applied to corruption data and can produce more robust estimates of economic costs. It will not rely on either method independently [4]. Studies also shown that the MIMIC model offers stronger structural interpretation. But it is less capable of adapting to rapidly changing informal sector dynamics. The ML models can be retrained more flexibly to capture evolving patterns [8][11].

Even though, the ML methods are superior in prediction and identifying complex patterns, the traditional econometric methods such as MIMIC and CDA have the ability to understand and explain the relationship between different factors like taxes, regulations, or unemployment influence the outcome. The ML models are also struggle to improve the performance accuracy when dealing with the small datasets (e.g., [8], [9], [10], [11]).

V. RESEARCH GAP AND FUTURE DIRECTION

The ML models outperform traditional econometric methods in estimating the informal economy size frequently, but there still exists some gaps. There are several factors affecting the measurement of informal economy, so the gaps depend on those factors. The main thing is, most of the research is carried out on single country or particular region. The next gap is related to the ML methods, there is no specific rules or methods for using machine learning when the data is small or low-quality, and in such situations, there is less chance for performance accuracy of the model. The ML models have high prediction accuracy but it shows low performance in causal estimation that affect the informal economy. The next gap identified from the research is that, the methods are not widely adopted the state-of-art deep learning architectures such as LSTMs and transformers. Also, it is not considering the new data sources like satellite imagery, mobile phone records etc.

Based on these research gaps there is a wide opportunity for future research. The developing of hybrid models that combine the advantages of both ML methods and traditional econometric methods is a novel research area. The possibility of doing research on multi country analysis on small-sample data is an important sector for future work. Incorporating the advanced ML and AI methods, using advanced data sources also interesting in this research area. The future research needs to have some of the following prioritize like hybrid model developments and causally explainable AI. Also, it should be like, advanced reproducible field that could isolate comparisons in apt methodology.

VI. CONCLUSION

This review concludes that the ML models consistently enhance the traditional econometric methods like MIMIC and CDA. Through the best predictive accuracy using non-linear relationship handling and integration of mixed-data sources, we could estimate the size of the informal economy. The ML models helps to lower out-of-sample prediction errors when compared to MIMIC and CDA. Since they use the validation metrics like RMSE and MAE [1], [2], [5]. Advanced approaches further outperform traditional methods by directly modelling spatial dependencies, such as estimating a country's informal economy based on neighbouring country characteristics [3], [6]. That said, traditional econometric methods still hold the edge when it comes to causal inference and structural interpretation. They let researchers actually test hypotheses about what drives informality, whether tax pressure, regulatory quality or any other factors. ML models work differently, built around prediction, they offer little transparency into why they produce the outputs. This makes it hard to illustrate actionable policy conclusions from them. Additionally, ML methods risk overfitting, whereas traditional methods rely on strong assumptions about currency velocity and baseline periods [8], [14], [15]. The review highlights hybrid approaches that combine the structural identification strengths of the MIMIC model with the residual prediction capabilities of ML as an emerging best practice in informal economy estimation [4]. The gaps related to different factors such as multi-country comparison, ML methods for small-sample datasets, neglected use of temporal deep learning architectures like LSTMs, and unused new data sources such as satellite imagery and mobile metadata. Future research should focus on implementation of hybrid models, use of explainable AI and novel data sources.

VII. AUTHOR CONTRIBUTION

The authors confirm the following contributions: study conception and topic analysis were carried out by Jinsy and Neethu; data collection, screening, and analysis were conducted by Jinsy and Neethu; and interpretation of results, along with manuscript writing, was completed by Jinsy and Neethu. Both authors reviewed and approved the final version of the manuscript.

REFERENCES

- [1] J. Felix, M. Alexandre, and G. T. Lima, "Applying machine learning algorithms to predict the size of the informal economy," *Computational Economics*, vol. 65, pp. 1169–1189, 2025.
- [2] A. D. Ivan, A. A. M. Davidescu, M. D. Agafitei, and M. C. Geambaşu, "Applying machine learning techniques to estimate the size of the Romanian shadow economy," in *Proc. Int. Conf. Business Excellence*, 2025, pp. 1–12.
- [3] L. Shami and T. Lazebnik, "Implementing machine learning methods in estimating the size of the non-observed economy," *Computational Economics*, vol. 63, pp. 1459–1476, 2024.
- [4] A. M. M. Saeed, M. A. Husnain, and M. Ali, "Quantifying the economic costs of financial corruption in Pakistan: An integrated econometric and machine learning approach," *Economies*, vol. 14, no. 3, art. no. 82, Mar. 2026.
- [5] T. Lazebnik, "Going a step deeper down the rabbit hole: Deep learning model to measure the size of the unregistered economy activity," *Computational Economics*, vol. 65, no. 3, pp. 1759–1774, Mar. 2025.
- [6] T. Lazebnik, L. Shami, and E. Peretz-Andersson, "Tell me who your neighbors are and I will tell you your informal economy size: The case of Sweden," *Computational Economics*, 2025, doi: 10.1007/s10614-025-11276-6.
- [7] A. S. Bennihi, L. Bouriche, and F. Schneider, "The informal economy in Algeria: New insights using the MIMIC approach and the interaction with the formal economy," *Economic Analysis and Policy*, vol. 72, pp. 470–491, 2021.
- [8] N. T. Hoa, "How large is Vietnam's informal economy?" *Economic Affairs*, vol. 39, pp. 81–100, 2019.
- [9] R. Machado, "The informal economy in Peru: Magnitude and determinants, 1980–2011," *Revista de Economía del Pacífico*, vol. XLI, no. 74, pp. 191–227, 2014.
- [10] R. Dell'Anno and O. Adu, "The size of the informal economy in Nigeria: A structural equation approach," *International Journal of Social Economics*, vol. 47, no. 8, pp. 1063–1078, 2020.
- [11] A. Asllani, N. Tzivanakis, and F. Schneider, "Unveiling the shadows: Tracing the informal economy in the Balkans from 1996 to 2021," *Post-Communist Economies*, vol. 37, no. 8, pp. 1093–1124, 2025.
- [12] K. S. Mughal, F. G. Schneider, and Z. Hayat, "Intensity of regulations as a cause of the informal sector," *Journal of South Asian Development*, vol. 15, no. 2, pp. 135–154, Aug. 2020.
- [13] M. Kiani, A. Ahmed, and K. Zaman, "Combining qualitative and quantitative approaches for measuring underground economy of Pakistan," *Quality & Quantity*, vol. 49, pp. 295–317, 2015.
- [14] S. Cichocki and A. Torój, "Estimating the size of informal economy in a post-transition country – The case of Poland," *Baltic Journal of Economics*, vol. 23, no. 1, pp. 91–116, 2023.
- [15] J. Atanasijević, M. Danon, Z. Lužanin, and D. Kovačević, "Shadow economy estimation using cash demand approach: The case of Serbia," *Sustainability*, vol. 14, no. 20, art. no. 13179, Oct. 2022.
- [16] B. Kitchenham, "Procedures for performing systematic reviews," Keele Univ., Keele, UK, Tech. Rep. TR/SE-0401, Jul. 2004.

List of Abbreviations

Abbreviations	Full Form
AI	Artificial Intelligence
CDA	Currency Demand Approach
CFI	Comparative Fit Index
GDP	Gross Domestic Product
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MIMIC	Multiple Indicators Multiple Causes
ML	Machine Learning
RMSE	Root Mean Square Error
RMSEA	Root Mean Square Error of Approximation
RQ	Research Question
SLR	Systematic Literature Review
SRMR	Standardized Root Mean Square Residual
SVM	Support Vector Machine
SVR	Support Vector Regression